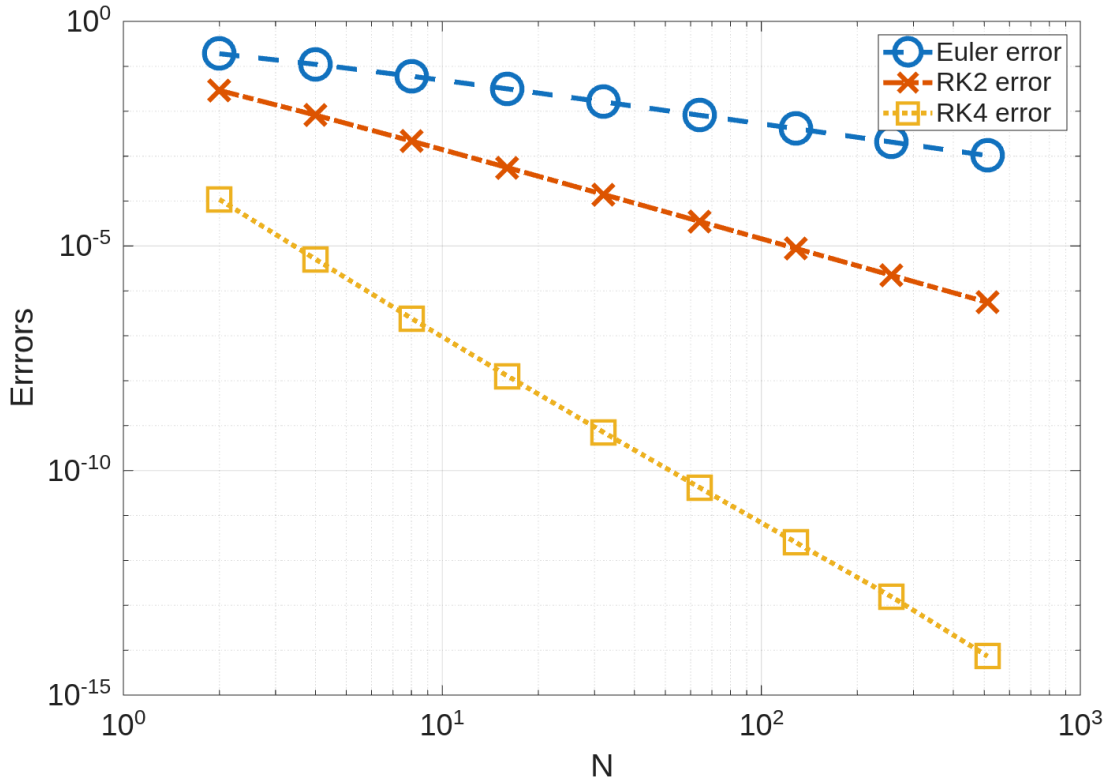
MA385 Part 2: Initial Value Problems

2.5: Runge-Kutta 4 (RK4)

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19 October 2026



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1 A higher order method

2 Consistency and convergence of RK4

3 The (Butcher) Tableau

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For more details, see Chapter 6 of Süli and Mayers, *An Introduction to Numerical Analysis*. In particular, see Section 12.4 (Runge-Kutta methods)

1. A higher order method

It is possible to construct methods that have higher orders of accuracy than **RK2** methods. Of these, the most used are probably those that belong to the **Runge-Kutta 4 (RK4)** family, and have the property that

$$|y(t_N) - y_N| \leq Ch^4.$$

However, even writing down the general form of the RK4 method, and then deriving conditions on the parameters is rather complicated. Therefore, we'll focus on just one RK4 method, and use examples, rather than theory, to demonstrate that it is 4th-order.

"The RK4 Method"

$$k_1 = f(t_i, y_i), \longrightarrow \text{Basic Euler.}$$

$$k_2 = f\left(t_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right), \longrightarrow \text{Modified Euler}$$

$$k_3 = f\left(t_i + \frac{h}{2}, y_i + \frac{h}{2}k_2\right),$$

$$k_4 = f(t_i + h, y_i + hk_3),$$

$$y_{i+1} = y_i + h\underbrace{\frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)}_{\Phi(t_i, y_i; h)} \quad \text{for } i = 0, 1, \ldots, N-1.$$

1. A higher order method

The RK4 method can be interpreted as follows :

- $k_1 = f(t_i, y_i)$ - just Euler's Method.
- $k_2 = f(t_i + \frac{h}{2}, y_i + \frac{h}{2}k_1)$ - this is the Modified RK2 from Section 2.4: error should be $\mathcal{O}(h^2)$.
- $k_3 = f(t_i + \frac{h}{2}, y_i + \frac{h}{2}k_2)$ - the Modified method, but using k_2. Error should be $\mathcal{O}(h^3)$.
- $k_4 = f(t_i + h, y_i + hk_3)$: using k_3.
- $y_{i+1} = y_i + h\dfrac{k_1 + 2k_2 + 2k_3 + k_4}{6}$: this is a "(1-2-2-1)/6" weighted average.

1. A higher order method

As the following example shows, **RK4** can be much more accurate than the Euler or RK2 methods for small h (i.e., large N).

For the **RK4**, halving h should scale the error by $(1/2)^4 = 1/16$.

By contrast, doubling N in an RK2 method would reduce the error only by a factor of 4.
For the basic Euler method, it would reduce the error by a factor of 2.

Example 2.5.1

Compare Euler (aka RK1), the Modified Euler RK2 method, and RK4 for approximating $y(1)$ *where:* $y(0) = 1$, *and* $y'(t) = y\log(1+t^2)$ *for* $t > 0$.

Error: $|y(t_N) - y_N|$

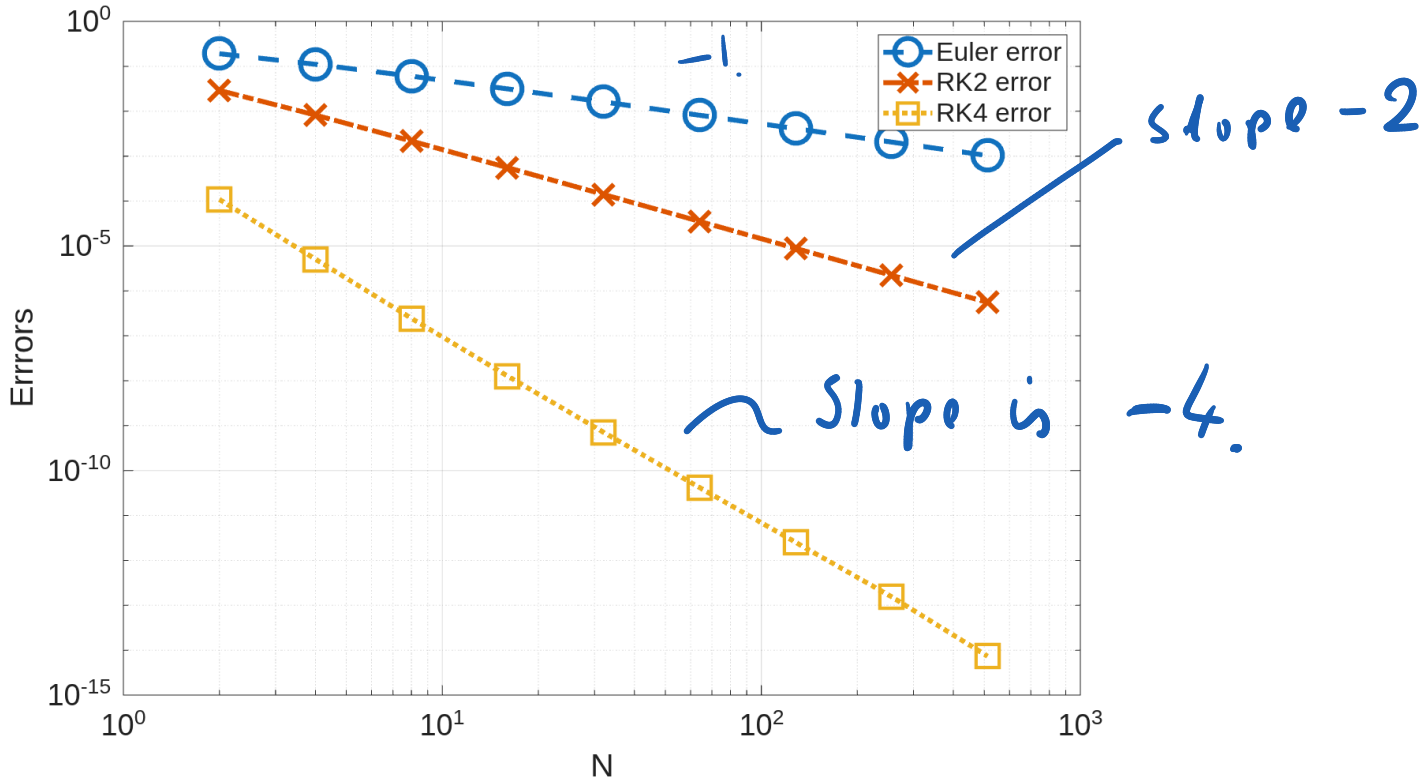
N	RK1	RK2	RK4
2	1.905e-01	2.901e-02	1.084e-04
4	1.107e-01	8.204e-03	5.073e-06
8	6.016e-02	2.164e-03	2.442e-07
16	3.144e-02	5.546e-04	1.271e-08
32	1.608e-02	1.403e-04	7.109e-10
64	8.135e-03	3.528e-05	4.176e-11
128	4.091e-03	8.844e-06	2.526e-12
256	2.052e-03	2.214e-06	1.543e-13
512	1.027e-03	5.539e-07	7.327e-15

Ratio ~ 16.

Example 2.5.2

Compare Euler (aka RK1), the Modified Euler RK2 method, and RK4 for approximating $y(1)$ *where:* $y(0) = 1$, *and* $y'(t) = y \log(1 + t^2)$ *for* $t > 0$.

log-log plot.



2. Consistency and convergence of RK4

Although we won't do a detailed analysis of RK4, we can do a little. In particular, we would like to show it is

(i) consistent,

(ii) convergent and fourth-order, at least for some examples.

Example 2.5.3

It is easy to see that RK4 is consistent. That is, that $\Phi(t_i, y_i; 0) = f(t_i, y_i)$.

With $h = 0$ we get

$$k_1 = k_2 = k_3 = k_4 = f(t_i, y_i)$$

Therefore

$$\Phi(t_i, y_i; 0) = \underbrace{\left(\frac{1}{6} + \frac{2}{6} + \frac{2}{6} + \frac{1}{6}\right)}_{1} f(t_i, y_i)$$

✓

Example 2.5.4

In general, showing the rate of convergence is tricky. Instead, we'll demonstrate how the method relates to a Taylor Series expansion for the problem $y' = \lambda y$ where λ is a constant. $\lambda \neq 0$

Since $\frac{dy}{dt} = \lambda y$ we have $y(t) = e^{\lambda t}$.

Note $y''(t) = \lambda^2 e^t$ $y'''(t) = \lambda^3 e^t$ etc.

So, in general,

$$y^{(n)}(t) = \lambda^n y(t)$$

2. Consistency and convergence of RK4

Now write out the Taylor Series for $y(t_{i+1})$ about $y(t_i)$. $h = t_{i+1} - t_i$

$$y(t_{i+1}) = y(t_i) + h\,y'(t_i) + \frac{h^2}{2}y''(t_i) + \frac{h^3}{3!}y'''(t_i) + \frac{h^4}{4!}y^{(4)}(t_i) + \frac{h^5}{5!}y^{(5)}(\eta)$$

some $\eta \in [t_i, t_{i+1}]$

So

$$y(t_i) = y(t_i)\left[1 + h\lambda + \frac{h^2\lambda^2}{2} + \frac{h^3\lambda^3}{6} + \frac{h^4\lambda^4}{24}\right] + \mathcal{O}(h^5)$$

2. Consistency and convergence of RK4

Next, write out the RK4 method for this problem.

$$k_1 = f(t_i, y_i) = \lambda y_i$$

$$k_2 = f(t_i + h/2,\ y_i + \tfrac{h}{2} k_1) = \lambda\left(y_i + \tfrac{h}{2}\lambda y_i\right) = y_i\left(\lambda + \tfrac{h\lambda^2}{2}\right)$$

$$k_3 = f(t_i + h/2,\ y_i + \tfrac{h}{2} k_2) = \lambda\left(y_i + \tfrac{h}{2} y_i\left(\lambda + \tfrac{h\lambda^2}{2}\right)\right)$$

$$= y_i\left(\lambda + \frac{h\lambda^2}{2} + \frac{h^2\lambda^3}{4}\right)$$

$$k_4 = f(t_i + h,\ y_i + h k_3) = \lambda\left(y_i + h y_i\left(\lambda + \frac{h\lambda^2}{2} + \frac{h^2\lambda^3}{4}\right)\right)$$

$$= y_i\left(\lambda + h\lambda^2 + \frac{h^2\lambda^3}{2} + \frac{h^3\lambda^4}{4}\right)$$

And then the method is

$$y_{i+1} = y_i + \frac{h}{6}k_1 + \frac{h}{3}k_2 + \frac{h}{3}k_3 + \frac{h}{6}k_4 .$$

2. Consistency and convergence of RK4

Now compare the Taylor Series and the method, matching powers of λ. (This bit was done on the board in class, and added here after). Write out RK4, collecting powers of λ:

$$y_{i+1} = y_i + \frac{h}{6}(\underbrace{\lambda y_i}_{k_1}) + \frac{h}{3} y_i \underbrace{\left(\lambda + \frac{h\lambda^2}{2}\right)}_{k_2}$$

$$+ \frac{h}{3} y_i \underbrace{\left(\lambda + \frac{h\lambda^2}{2} + \frac{h^2\lambda^3}{4}\right)}_{k_3} + \frac{h}{6} y_i \underbrace{\left(\lambda + h\lambda^2 + \frac{h^2\lambda^3}{2} + \frac{h^3\lambda^4}{4}\right)}_{k_4}$$

$$= y_i\left(1 + \lambda\underbrace{\left(\frac{h}{6} + \frac{h}{3} + \frac{h}{3} + \frac{h}{6}\right)}_{1} + \lambda^2\underbrace{\left(\frac{h^2}{6} + \frac{h^2}{6} + \frac{h^2}{6}\right)}_{1/2} + \lambda^3\underbrace{\left(\frac{h^3}{12} + \frac{h^3}{12}\right)}_{1/6} + \lambda^4\left(\frac{h^4}{24}\right)\right)$$

$$= y_i\left(1 + h\lambda + \tfrac{1}{2}h^2\lambda^2 + \tfrac{1}{6}h^3\lambda^3 + \tfrac{1}{24}h^4\lambda^4\right).$$ This is the same as the Taylor Series on Slide 11, up to, and including h^4 terms.

3. The (Butcher) Tableau

Many (seemingly different) RK have been proposed and studied. A unified approach of representing them was developed by John Butcher: write an s-stage method as

$$\Phi(t_i, y_i; h) = \sum_{j=1}^{s} b_j k_j, \qquad \text{where}$$

$$\begin{aligned}
k_1 &= f(t_i + \alpha_1 h, y_i), \\
k_2 &= f(t_i + \alpha_2 h, y_i + \beta_{21} h k_1), \\
k_3 &= f(t_i + \alpha_3 h, y_i + \beta_{31} h k_1 + \beta_{32} h k_2), \\
&\vdots \\
k_s &= f(t_i + \alpha_s h, y_i + \beta_{s1} h k_1 + \ldots \beta_{s,s-1} h k_{s-1}),
\end{aligned}$$

3. The (Butcher) Tableau

The most convenient way to represent the coefficients is in a tableau:

$$
\begin{array}{c|cccccc}
\alpha_1 & & & & & & \\
\alpha_2 & \beta_{21} & & & & & \\
\alpha_3 & \beta_{31} & \beta_{32} & & & & \\
\vdots & & & & & & \\
\alpha_s & \beta_{s1} & \beta_{s2} & \cdots & \beta_{s,s-1} & & \\
\hline
 & b_1 & b_2 & \cdots & b_{s-1} & b_s
\end{array}
$$

The tableaux for basic Euler, Modified Euler, and RK4 are:

$$
\begin{array}{c|cc}
0 & & \\
\hline
 & 1 &
\end{array}
\qquad
\begin{array}{c|cc}
0 & & \\
1/2 & 1/2 & \\
\hline
 & 0 & 1
\end{array}
\qquad
\begin{array}{c|cccc}
0 & & & & \\
1/2 & 1/2 & & & \\
1/2 & 0 & 1/2 & & \\
1 & 0 & 0 & 1 & \\
\hline
 & 1/6 & 2/6 & 2/6 & 1/6
\end{array}
$$

4. Even higher-order methods?

A Runge Kutta method has s stages if it involves s evaluations of the function f. (That it, its formula features $k_1, k_2, \ldots, k_s$).

- We've seen a 1-stage method that is 1st-order.
- We studied 2-stage methods that are 2nd-order.
- As a tutorial exercise, you'll construct a 3-stage method that is 3rd order.
- And, of course, we have just considered a four-stage method that is 4th-order.

4. Even higher-order methods?

It is tempting to think that we could we can get an Order s method by using s stages for any s.
However, it can be shown that:

- for a method to have order $p = 5$, we need $s = 6$ stages (Kutta, 1901)
- for order $p = 6$, we need at least $s = 7$ stages (Butcher, 1964).
- for order $p = 7$, we need at $s = 9$ stages (Butcher, 1968).
- for order $p = 8$, we need at $s = 11$ stages (Curtis, 1970).
- for order $p = 10$, we need at $s = 17$ stages (Hairer, 1978).

The theory involved is both intricate and intriguing, and involves aspects of group theory, graph theory, and differential equations.
See https://doi.org/10.1016/0168-9274(95)00108-5

5. Exercises

Exercise 2.5.1

We claim that, for *RK4*:

$$|\mathcal{E}_N| = |y(t_N) - y_N| \leq Kh^4.$$

for some constant K. How could you verify that the statement is true using the data of Table 2.3, at least for test problem in Example 2.4.2? Give an estimate for K.

5. Exercises

Exercise 2.5.2

Recall the problem in Example 2.2.2: *Estimate* $y(2)$ *given that*

$$y(1) = 1, \qquad y' = f(t, y) := 1 + t + \frac{y}{t},$$

(i) Show that $f(t, y)$ satisfies a Lipschitz condition and give an upper bound for L.

(ii) Use Euler's method with $h = 1/4$ to estimate $y(2)$. Using the true solution, calculate the error.

(iii) Repeat this for the RK2 method of your choice (with $a \neq 0$) taking $h = 1/2$.

(iv) Use *RK4* with $h = 1$ to estimate $y(2)$.

5. Exercises

Exercise 2.5.3

Here is the tableau for a three stage Runge-Kutta method:

0			
α_2	1/2		
1	β_{31}	2	
	1/6	b_2	1/6

(i) Use that the method is consistent to determine b_2.

(ii) The method is exact when used to compute the solution to

$$y(0) = 0, \quad y'(t) = 2t, \ t > 0.$$

Use this to determine α_2.

(iii) The method should agree with an appropriate Taylor series for the solution to $y'(t) = \lambda y(t)$, up to terms that are $\mathcal{O}(h^3)$. Use this to determine β_{31}.